Applications of Derivatives (AOD)

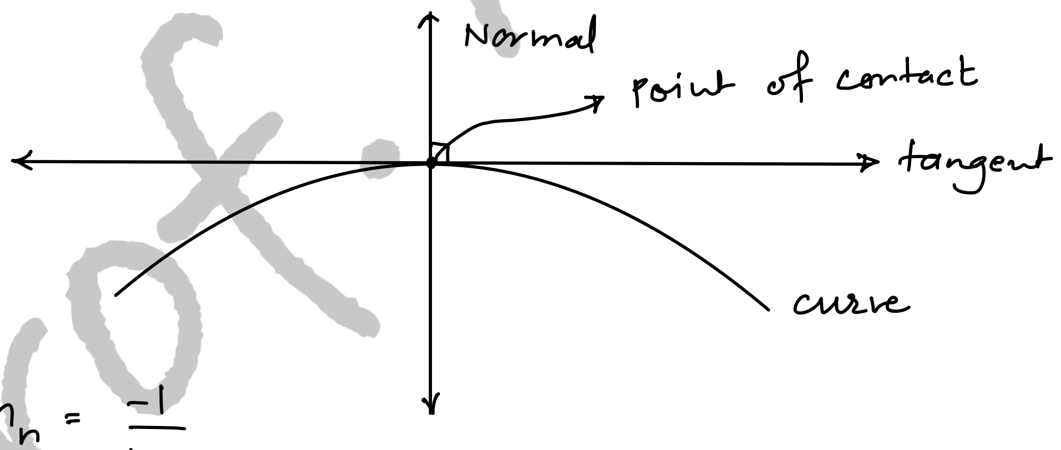
Types of sums :

1) Tangents & normals
2) Rates
3) Displacement, velocity, acceleration
4) Approximate values
5) Increasing, decreasing functions
6) Maxima & Minima
7) Rolle's Theo^m & LMVT.

Tangents & normals

① slope of tangent $m_t = \frac{dy}{dx}$ at (x_1, y_1)

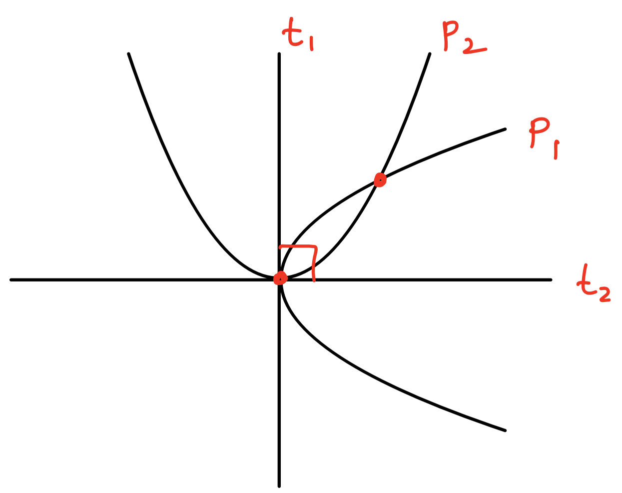
② slope-point form : $y - y_1 = m(x - x_1)$



③ $m_n = \frac{-1}{m_t}$

Orthogonal curves :

Two curves are orthogonal if tangents at their point of intersection are $\perp$



Displacement, velocity, acceleration

displacement : s

velocity : $v = \frac{ds}{dt}$

acc^n : $a = \frac{dv}{dt}$ or $\frac{d^2s}{dt^2}$

avg. vel. = $\frac{\text{total dist travelled}}{\text{total time}}$

Approximate values

$$\boxed{f(a \pm h) = f(a) \pm h f'(a)}$$ — [h : very small]

Increasing, decreasing functions :

- ★ If $f'(x) \geq 0$ then $f(x)$ is increasing f^n
- ★ If $f'(x) \leq 0$ then $f(x)$ is decreasing f^n
- ★ If $f'(x) > 0$ then $f(x)$ is strictly / monotonically incrasing f^n
- ★ If $f'(x) < 0$ then $f(x)$ is strictly / monotonically decreasing f^n

Maxima & Minima

Steps :

1) Find $f'(x)$
2) Find $f''(x)$
3) Put $f'(x) = 0$ & find $x_1, x_2, \ldots\ldots$
4) Find $f''(x_1)$

   If $f''(x_1) < 0$ then $f(x)$ is maximum, find f_{max}

   If $f''(x_1) > 0$ then $f(x)$ is minimum, find f_{min}
5) Repeat step-4 for $x_2, \ldots\ldots$